

The Grid Fixed Itself. Transport Didn't.

UK Greenhouse Gas Emissions 1990-2024 — Bootstrap CUSUM Analysis — Executive Summary
stepchangeanalysis.com | [DESNZ & DEFRA Data](#) | [May 2026](#) | [Syd Stewart, Chartered Chemical Engineer](#)

Note: Territorial emissions from DESNZ Final UK GHG Statistics 1990-2024, N=35 annual observations. Consumption-based emissions from DEFRA UK Carbon Footprint to 2022, N=27. Bootstrap CUSUM, Loops=10,000, Turn Length=5, 90% confidence unless stated.

Five Key Findings

- 1

The 53% headline reduction has three structural causes — only one is climate policy

Bootstrap CUSUM identifies four structural stages in UK territorial emissions 1990-2024. Stage 1-2 (1998): -8.4% at 95.46% — dash for gas, energy market liberalisation, not climate policy. Stage 2-3 (2008): -17.2% at 93.88% — global financial crisis, GDP fell 4.2%. Stage 3-4 (2015): -24.5% at 93.88% — coal phase-out and renewables, the first genuinely policy-driven stage. The majority of structural reduction is in the most recent stage — arriving after 25 years, not progressive policy since 1990.

N=35 | Stages=4 at 90% | Stages=2 at 99.7% | Only 2008 beyond reasonable doubt
- 2

Electricity: 23 years of stagnation, then one policy changed everything

UK electricity supply emissions were essentially flat from 1990 to 2013 — despite Kyoto, the Renewables Obligation, and the Climate Change Act. The carbon price floor (April 2013) changed the economics of coal generation overnight. Result: -55.4% in 11 years at 99.8% confidence — the strongest signal in the entire dataset. The renewables that had been building for a decade were ready to fill the gap. The unintended consequence worked because it changed the system, not individual behaviour.

Stage 1: 1990-2013, mean 167.53 MtCO₂e | Stage 2: 2013-2024, mean 74.76 MtCO₂e | Conf: 99.8%
- 3

Transport: no structural change at 95% confidence across 34 years

Domestic transport is the largest remaining sector at 30% of all UK emissions. Bootstrap CUSUM finds four stages at 90% confidence — but all four boundaries fall below 95%. The UK population grew 19% from 57M to 68M in the same period. Net absolute reduction: 10.6%. Stage 4 (2019-2024) is partly a COVID artefact — lockdowns collapsed transport activity in 2020. The sector has not structurally decarbonised.

Stage 2: 93.24% | Stages 3-4: 91.85% | All below 95% threshold | Population +19%
- 4

EVs are not yet visible in the data — and the maths explains exactly when they will be

1.2M EVs out of 35M vehicles = 3.5% of the fleet. Annual saving approximately 1.44 MtCO₂e against a sector total of 108 MtCO₂e — 1.3% of emissions, below the statistical noise floor. To produce a detectable Bootstrap CUSUM change point requires 12-15M EVs on the road. At 300,000-400,000 new EVs/year, that fleet share arrives around 2034-2038. The Bootstrap CUSUM is not a verdict on EV policy. It is a clock.

Detection threshold: ~12-15M EVs | Expected change point: 2034-2038 | Current fleet share: 3.5%
- 5

Consumption-based emissions confirm 2008 and 2015 are real — but 1998 is not

DEFRA consumption-based emissions (N=27, 1996-2022) show the same change points at 2008 and 2015 as the territorial series — confirming those reductions are genuine on both measures. Crucially, the 1998 territorial stage change has NO consumption equivalent. That early -8.4% reduction reflects production restructuring (dash for gas), not genuine decarbonisation of UK economic activity. The offshoring signal is precisely located.

Consumption stages: 2008 (-17.8%) and 2015 (-23.4%) | No 1998 equivalent | Both below 95%

The Systems Thinking Argument

Framework	Applied to UK Carbon Emissions
Reinforcing loop R1 (electricity)	Carbon price floor raised coal costs -> closures accelerated -> renewables filled the gap -> further reduced coal viability. No compensating mechanism existed — coal operators could not make coal cheap again. The enabling conditions (EU ETS collapse, cheap gas, decade of renewable build) were already in place. Result: 99.8% confidence change point within two years of policy taking effect.
Balancing loop B1 (transport)	Fuel duty raised driving costs -> vehicle efficiency improved -> cost per mile fell -> miles driven maintained. Population growth (+19%) absorbed further gains. The system absorbed the cost signal through the efficiency feedback loop. Result: no structural change at 95% confidence across 34 years.
Senge — policy resistance (5th Discipline, 1990)	Well-intentioned interventions trigger compensating feedback loops that defeat the solution. Transport fuel duty faces exactly this resistance. The carbon price floor escaped it because it worked with the market feedback loop rather than against it — making the desired outcome the economically rational choice for every generator simultaneously.
Deming — system vs individual (Out of the Crisis, 1982)	Electricity: the carbon price floor changed the system. No individual behaviour change required. Transport: fuel duty, emission zones, and vehicle tax bands apply pressure to individual decisions within an unchanged system. 35 million vehicle owners make individual choices in a system designed around fossil fuel combustion. Until the system is redesigned, transport will continue to produce marginal signals below 95% confidence.
Policy lag	There is always a lag between policy and detectable outcome. The carbon price floor had a short lag (~2 years) because it acted on a centralised system of 300 power stations. Transport policy has a structural lag of 10-15 years because fleet turnover takes that long. The EV transition policies (2020 onwards) are within the lag window — the CUSUM cannot yet say they have failed. Expected change point: 2034-2038.

What Bootstrap CUSUM Tells Policymakers

Sector	Policy mechanism	CUSUM verdict	Why it worked / didn't
Electricity	Carbon price floor 2013	99.8% confidence -55.4% in 11 years	Changed system economics for all 300 generators simultaneously. No compensating loop.
Transport	Fuel duty, emission zones, vehicle tax 1990-2024	No change at 95% confidence	Applied pressure to 35M individual decisions. Efficiency gains absorbed the cost signal.
Transport (prospective)	EV mandate, 2030 ban, charging infrastructure	Within lag window Expected 2034-2038	Fleet turnover takes 15 years. Too early to assess. CUSUM will detect change point objectively.



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Full interactive analysis, all charts, Bootstrap CUSUM tool and data downloads:

stepchangeanalysis.com/the-grid-fixed-itself.html

Key References

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DEFRA — UK Carbon Footprint to 2022 (May 2025)
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